



Antibacterial-renew dual-function anti-biofouling strategy: Self-assembled Schiff-base metal complex coatings built from natural products

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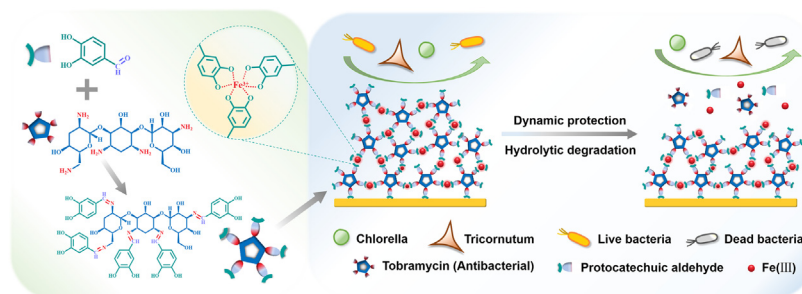
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HIGHLIGHTS

- Self-assembled Schiff-Base metal complex coatings were built from completely natural products.
- The coating degradation was uniform and controlled and would not led to the accumulation of microplastics.
- (Fe/TOB-PR)₂₀ coating exhibited excellent antimicrobial properties and resistance to microbial adhesion.
- The coating could be assembled on various substrates.

GRAPHICAL ABSTRACT

We prepared a new environment-friendly dual-function anti-fouling coating with simultaneous self-renewal and sterilization. The (Fe/TOB-PR)₂₀ coating displayed excellent antimicrobial properties and resistance to microbial adhesion which was attributed to the renewal of the surface and the generation of antibacterial substance (TOB) in the coating.



ARTICLE INFO

Article history:

Received 14 June 2022

Revised 26 August 2022

Accepted 29 August 2022

Available online 5 September 2022

Keywords:

Anti-biofouling coating

Schiff base

Natural modules

Synergistic effect

Self-renewable coating

Antibacterial

ABSTRACT

Marine biological fouling has caused huge economic losses and environmental problems. Therefore, it was essential to develop effectively environment-friendly biofouling resistance technology. Here, inspired by the natural module of bacterial secretions, animal and plant extracts, we synthesized Schiff based compounds through Tobramycin (TOB) from *Streptomyces* and Protocatechualdehyde (PR) from black fern. Furthermore, a dynamic self-renewal Schiff based metal composite coating- (Fe/TOB-PR)_n was prepared via layer by layer self-assembly (LBL) method. It was proved to be a versatile coating, which could adhere to different types of substrates. Hydrolytic degradation tests showed that the degradation of the (Fe/TOB-PR)₂₀ coating was regular and controlled. Moreover, compared with the blank glass substrate, the antibacterial rate of (Fe/TOB-PR)₂₀ reached 97 % after 24 h, and the test further shows that the durability of the antibacterial performance benefits from the greater coating thickness. Such coatings displayed excellent anti-bacteria and anti-algae adhesion properties which was attributed to the renewal of the surface and the generation of antibacterial substance (TOB) in the coating. Further, the coating eventually degraded to natural micromolecule monomer, avoiding the occurrence of microplastics. It provides research ideas for fabricating environment-friendly anti-biofouling coatings.

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1. Introduction

In the marine industry, the adhesion, growth and reproduction of microorganisms and algae on the surface of underwater equipment will promote the attachment of other marine organisms and accelerate the corrosion of metal materials, resulting in excessive energy consumption and maintenance costs [1–4]. At present, anti-biofouling coating was considered as the most convenient and widespread method to protect marine equipment. After the traditional organotin-containing antifouling coating was banned in 2003, the antifouling coating containing Cu^{2+} was widely used. Now, the concentration of Cu^{2+} in some sea areas has increased, which was potentially harmful to the environment and human [5,6]. Obviously, copper-containing antifouling coatings are at risk of being banned.

In order to protect the marine environment and maintain the marine ecological balance, more and more attention has been paid to the research of nontoxic self-polishing coatings, degradable coatings, low surface energy coatings and biomimetic coatings containing natural active substances [7–10]. Among them, the difficult-to-degrade polymer backbones in nontoxic self-polishing antifouling coating may lead to the enrichment of microplastics in marine organisms [11,12]. In contrast, the side chain of polymer in degradable coating could be completely hydrolyzed, and its main chain could be decomposed by the enzymatic action of microorganisms (such as bacteria, fungi and algae), thus reducing the release and aggregation of microplastics. It not only meets the requirement of environmental protection, but also is expected to solve the problem of poor antifouling capability of self-polishing coating under static condition [13–15]. Xu et al., researchers from the South China University of Technology, has prepared the main chain degradation-side chain hydrolysis polyurethane and the main chain degradation-side chain hydrolysis polyacrylate for preventing biofouling. The hydrolysis performance of the side chain silane ester was effectively combined with the degradation performance of the main chain, and its excellent antifouling ability was proved by the marine field antifouling test [16]. Integrating the fouling-release or degradable coatings with biofriendly antifouling agents to improve antifouling performance is an effective strategy. Ma et al. [17] combined degradable Polycaprolactone (PCL) with 5-octylfuran-2(5H)-1 (a butenolide)(an antibacterial agent originated from marine *Streptomyces*), which improved the antifouling ability of the polymer and prolonged the service life. Therefore, degradable self-renewal surface has excellent antifouling performance, which is a good strategic direction to solve the urgent need of marine antifouling [8]. At present, the most basic units of degradable antifouling materials are almost all derived from synthetic chemicals, which still pose a potential risk to the ecological environment. It is still difficult to design and fabricate a kind of self-renewal antifouling coating, which simultaneously possess all-natural antibacterial, anti-bioadhesive and environment-friendly properties.

Antifouling coatings constructed entirely from natural product modules are a forward-looking strategy to solve these issues. In recent years, cases of degradable coatings made from natural products are emerging, typically represented by tannins (TA) and dopamine (DA) [18–21]. For example, Hirotaka Ejima et al. [22] used polyphenol TA and Fe^{3+} as organic ligands and inorganic crosslinkers to form coordination complexes for one-step coating at various interfaces. Their work demonstrated an extremely rapid technique for the production of structurally diverse and decomposable films. The as-prepared coating contained no resin matrix, could be completely degraded, had low cost, simple assembly process. Suyeob Kim et al. [18] developed reversible layer-by-layer deposition by imitating the characteristics of mussel foot silk cuticle, using cate-

chol compounds and iron ions to form a complex to construct LBL films, and confirmed the degradability of the coating structure. Inspired by the above research, we want to construct a new type of completely degradable and environmental friendly coating for marine antifouling application, and the degradation products were completely natural products. Noticeably, the Schiff bases were formed by the condensation of carbonyl and amine groups, which could coordinate with metal ions effectively. Moreover, the synthesis method of Schiff base was simple, and could be hydrolyzed into aldehydes and amines, it has been reported as effective antibacterial active compounds [23,24]. In order to further improve the antibacterial properties of Schiff base compounds, it was known that the tobramycin (TOB) extracted from *Streptomyces*, with multiple amino groups and good hydrophilicity, has good antibacterial properties against Gram-negative and Gram-positive bacteria [25,26]. Besides, the protocatechualdehyde (PR) is a plant-derived phenolic aldehyde compound and containing an aldehyde group and two phenolic hydroxyl groups, which is broad-spectrum antibacterial agent [27,28]. The synthesis of Schiff base by TOB and PR to construct the coating not only enhances the antibacterial property of the system, but also meets the requirements of environment-friendly products.

In this work, a Schiff base compound (TOB-PR) was synthesized from protocatechualdehyde and tobramycin in a molar ratio of 1:5. Each monomer of the Schiff base had five catechol groups, which could be complexed with Fe^{3+} . Based on this principle, the modular layer-by-layer self-assembly was performed on the substrate surface, resulting in a firmly connected ultrathin Schiff base metal composite coating. The coating was composed entirely of natural products and could be applied to a variety of materials surfaces due to the tannic acid-like structure. Due to the hydrolysis characteristics of the imine bond, the Schiff base metal composite coating with a certain thickness are expected to exhibit controlled degradation capability, as well as provide intensified anti-biofouling capability since the release of natural anti-fouling degradation products. The degradation, antibacterial properties and anti-bioadhesion properties of Schiff base metal composite coating were evaluated and the biofouling resistance mechanism was analyzed. Therefore, a new completely degradable anti-biofouling coating was explored in this study, which may open up a new way for the design and preparation of novel antifouling coatings.

2. Materials and methods

2.1. Materials

Tobramycin ($\geq 98\%$, T10087-25 g) was purchased from Aladdin Co. Ltd. Protocatechualdehyde ($\geq 98\%$, D807224-100 g) was purchased from Macklin Biochemical Co., Ltd. Methanol (AR Grade, $\geq 99.5\%$), Ethanol absolute (AR Grade, $\geq 99.0\%$) and Anhydrous Ferric chloride (FeCl_3 , $\geq 98\%$) were purchased from Sinopharm Chemical Reagent Co., Ltd. Phosphate buffer saline (PBS buffer) was purchased from Sangon Biotechnology Co., Ltd. (Shanghai, China). *E. coli* (ATCC25922) was purchased from Shihai luwei Technology Co., Ltd. *Chlorella* sp. and *Phaeodactylum tricornutum* were purchased from Shanghai Guangyu Biological Technology Co., Ltd.

2.2. Synthesis of TOB-PR

Firstly, 2.7627 g of Protocatechualdehyde (PR) was accurately weighed and dissolved in 50 mL methanol, stirred thoroughly at a constant temperature of 60°C for 2 h. Then, 1.8701 g of Tobramycin (TOB) was dissolved in 80 mL methanol. After the completely dissolved TOB solution was slowly added into PR solution at the rate of one drop per second, the reaction mixture was continuously

stirred at 60 °C for 22 h. Secondly, the solvent was removed from the fully reacted mixture by rotary evaporation. The product was taken out and a certain amount of methanol was added to make the product fully contact with methanol, and then centrifugation were performed twice to purify it. Finally, the product was vacuum dried at 40 °C to obtain the Schiff base product TOB-PR (yellow–brown powder).

2.3. Assembly of $(\text{Fe}/\text{TOB-PR})_n$ coating

Through repeated experiments, the coatings' assembly steps were determined. Firstly, the glass substrate was ultrasonically cleaned with deionized water, ethanol and acetone for 10 min, and dried at 60 °C for standby. Secondly, 0.2 g/L of TOB-PR solution and 0.25 g/L of FeCl_3 solution solutions were stirred evenly, which was prepared with the methanol as solvent to add into two dyeing cylinders respectively. In addition, methanol was added to the other four dyeing cylinders (submerge the dyeing rack during cleaning). Finally, the coating of $(\text{Fe}/\text{TOB-PR})_n$ was assembled according to the following steps (as shown in Fig. 1B): (1) The substrate was immersed in 0.2 g/L TOB-PR solution for 25 min, quickly cleaned it twice with methanol solution to remove the excess TOB-PR, and then dried it naturally; (2) Soaked in 0.25 g/L FeCl_3 solution for 25 min, quickly cleaned twice with methanol solution, and then dried naturally. Repeated the above steps to assemble 1/2 (TOB-PR), 1, 3, 5, 7, 10, 15 and 20 layers of $(\text{Fe}/\text{TOB-PR})_n$ coating, respectively. The “n” represented repeating “n” times, meaning the number of layers.

2.4. Characterization of polymers TOB-PR

The FT-IR spectra of TOB, PR, and TOB-PR were recorded via a Nicolet 6700 FT-IR spectrometer (Thermo Scientific, America) in the 4000–500 cm^{-1} range. The UV–vis absorption spectra of the raw materials and products were determined by UV–vis near infrared spectrophotometer (Lambda 950, Perkin-Elmer, America). The ^1H nuclear magnetic resonance (^1H NMR) (AVANCE III 400 MHz, Bruker, Switzerland) of TOB, PR, and TOB-PR was recorded at room temperature using DMSO as solvent separately.

2.5. Characterization of $(\text{Fe}/\text{TOB-PR})_n$ coating

UV–vis absorption spectra of $(\text{Fe}/\text{TOB-PR})_{20}$ coatings assembled on different substrates and $(\text{Fe}/\text{TOB-PR})_n$ coatings was acquired with the UV near infrared spectrophotometer (Lambda 950, Perkin-Elmer, America). Except for the fact that different types of substrates were used for the above tests, the coatings used for other characterization and testing were all assembled on a common glass substrate. The water contact angle of $(\text{Fe}/\text{TOB-PR})_n$ coating was measured by a dynamic water contact angle measuring instrument (DCAT21, Dataphysics, Germany), each sample was tested at 6 different positions and then averaged.

To accurately measure the coating thickness, straight scratches (bare substrate) were firstly fabricated on the coating surface with a scalpel, and then the coating thickness was characterized using a scanning probe microscope (SPM, Dimension ICON, Bruker, America) at room temperature, and the coating morphology and surface

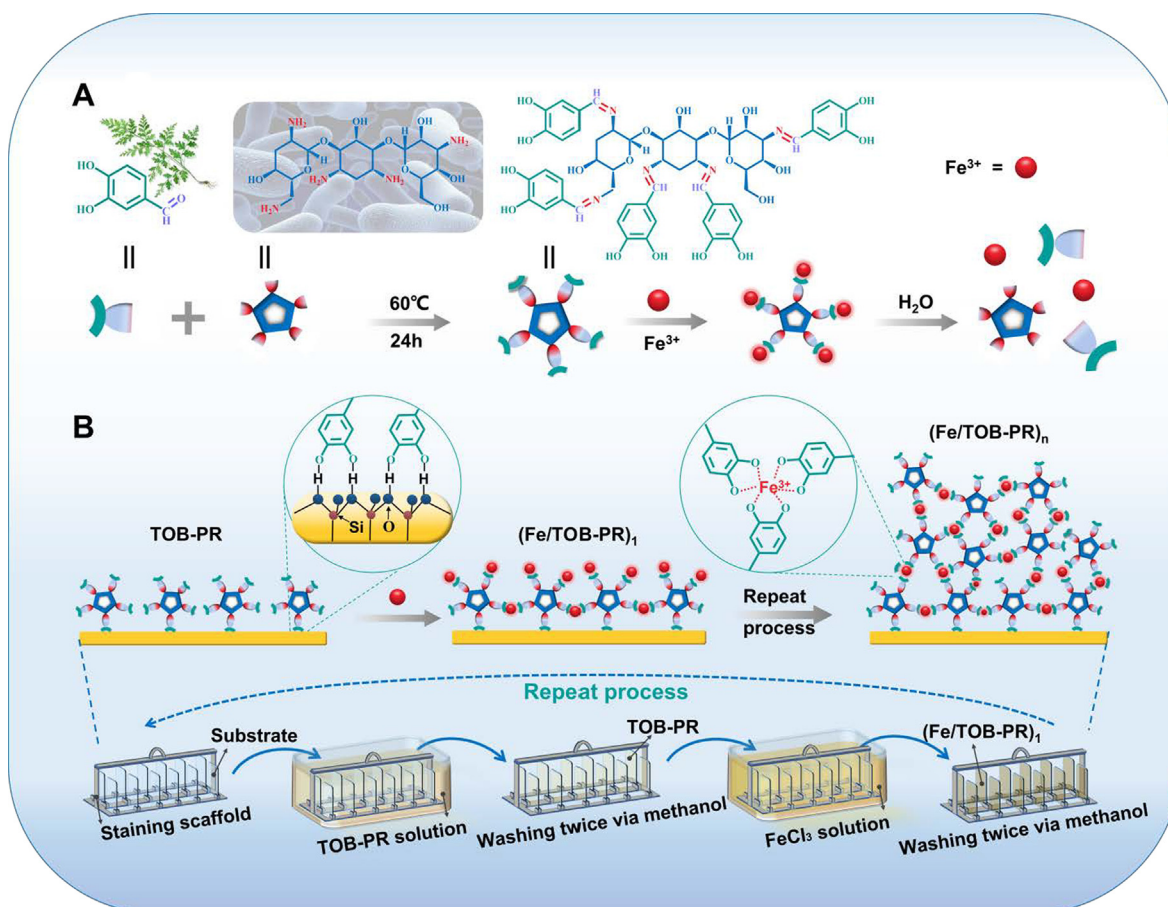


Fig. 1. Design concept of TOB-PR and Schiff base metal complexes. A) Synthesis of Schiff base compound TOB-PR and schematic diagram of coating assembly principle and its degradation principle. B) Ideal internal structure of $(\text{Fe}/\text{TOB-PR})_n$ coating based on the combination of TOB-PR and Fe^{3+} and schematic diagram of layer-by-layer self-assembly process of coating. The coordination between protocatechin and Fe^{3+} is the bridge for assembling the coating.

roughness were also characterized using the same instrument. The chemical composition of (Fe/TOB-PR)_n coating was detected with an X-ray photoelectron spectrometer (XPS, AXIS SUPRA, Kratos, England). To analyze the optimum complex ratio of catechol group and Fe³⁺, different molar ratios of TOB-PR: Fe³⁺ solution were prepared by methanol as solvent (1:0.5, 1:1, 1:1.25, 1:1.67, 1:2.5, 1:5 and 1:10, respectively. See Table S1 for details), which were characterized by UV near infrared spectrophotometer (Lambda 950, Perkin-Elmer, America).

2.6. Degradation test of (Fe/TOB-PR)₂₀ coating

(Fe/TOB-PR)₂₀ Coatings (76 mm × 26 mm) were statically placed in 45 mL deionized water and 45 mL artificial seawater at 25 °C, respectively. 2.5 mL of solution after immersion was taken out respectively at 2, 6, 12, 24, 48, 72, 96 and 120 h, the solution composition was measured by UV–vis spectrophotometer (Lambda 950, Perkin-Elmer, America). And the solution taken out after test needs to be poured back. In addition, another group of (Fe/TOB-PR)₂₀ coatings (76 mm × 26 mm) were statically immersed in 45 mL deionized water and 45 mL artificial seawater at 25 °C, which were taken out at 0, 12, 24, 36, 48, 72, 96 and 120 h respectively, and UV test was performed on the coating samples after dried naturally (room temperature, 48 h). Therefore, the degradation curves of (Fe/TOB-PR)₂₀ coating were obtained within 120 h. The above experiments were repeated more than three times. At the same time, the changes of coating thickness and roughness were recorded by SPM (Dimension ICON, Bruker, America). XPS (AXIS SUPRA, Kratos, England) fine spectra were used to analyze the chemical composition of the coating before and after 3 h and 72 h of degradation.

2.7. Antibacterial tests of (Fe/TOB-PR)_n coating

The antibacterial activity of the (Fe/TOB-PR)_n coating was evaluated used Gram-negative *Escherichia coli*. Firstly, the *E.coli* strain was inoculated into 50 mL LB broth, shaken at 128 rpm for 18 h in a constant temperature shaking incubator (MZQ-X100) at 37 °C. Subsequently, the LB broth was centrifuged at 3000 rpm for 5 min, and washed twice with PBS buffer to eliminate the residual culture medium. The obtained bacteria were dispersed in 50 mL PBS solution, and its concentration was determined by a microplate reader (MD SpectraMax 190, USA). Finally, PBS solution was used to dilute the cultured bacteria solution to 1.0×10^6 CFU/mL and 1.0×10^5 CFU/mL for further tests, respectively.

Inhibition zone test: First, 1280 µg/mL solution of TOB, PR, TOB-PR, and Fe/TOB-PR (TOB-PR had five catechol groups and the ratio of catechol to Fe³⁺ was 1:3 when fully complexed, so the Fe:TOB-PR molar ratio was 5:3) was prepared with methanol solvent. A circular filter paper piece with a diameter of 7 mm was immersed in the solution, and taken out for dried naturally (room temperature, 2 h) after 10 min. A pipettor gun sucked 100 µL of bacterial suspension with a concentration of 1.0×10^6 CFU/mL and uniformly coated on 2216E agar plates. The filter paper piece was placed on the agar plates coated with the bacterial suspension, and repeated three times. After being cultured in an incubator at 37 °C for 24 h, the diameter of an inhibition zone was measured with a ruler.

Spread plate method: (Fe/TOB-PR)_n (n = 1, 10, 20) coatings were respectively immersed in about 45 mL bacterial suspension (1.0×10^5 CFU/mL) as the control group, and bacterial solution soaked in blank glass was used as a reference. These bacterial solution (blank glass, (Fe/TOB-PR)_n (n = 1, 10, 20)) were kept in a constant temperature oscillation incubator (128 rpm, 37 °C), and 100 µL bacteria solution were taken out at 0, 6, 12, 18 and 24 h, and diluted to 10, 100 and 1000 times respectively. Then, take 100 µL of bacterial suspension by pipettor into 2216E agar plate,

and distribute the bacterial suspension evenly on the agar plate with spreader. After cultured in an incubator at 37 °C for 24 h, colony photographs were taken and counted using a colony counter. The bacteriostatic rate (Rs) was calculated by the following formula, where Ds represented the number of colonies in the blank glass and Dt represented the number of colonies in the (Fe/TOB-PR)_n coatings [25,29].

$$Rs = \frac{Ds - Dt}{Ds} \quad (1)$$

Live/dead bacterial cell assays: In order to explore the surface morphology of *Escherichia coli* before and after antibacterial test, the bacterial solution added with TOB-PR was used as the control group, two different concentrations of TOB-PR solution were set in the control group, namely 0.5 mg/mL and 64 µg/mL. Then the sample was immobilized by 2.5 % glutaraldehyde solution at 4 °C for 24 h, and dehydrated by 25 %, 50 %, 70 %, 80 %, 90 % and 100 % alcohol solution in turn. At last, the bacterial sediment was washed twice and dispersed with pure ethanol, and a small amount was dropped on the slide glass. After natural drying, the morphology of bacteria was observed by SEM (S4800, Hitachi, Japan).

2.8. Microorganisms adhesion tests of (Fe/TOB-PR)_n coating

Firstly, *E. coli* strain was inoculated into 50 mL sterile LB medium and cultured for 18 h at constant temperature incubator (128 rpm, 37 °C), and the concentration of the obtained bacterial solution to achieve greater than 1.0×10^8 CFU/mL by a microplate reader. The (Fe/TOB-PR)₂₀ coating and blank glass (26 mm × 26 mm) were placed flat in a 6-well plate with 10 mL bacterial solution and shaken at a constant temperature incubator (128 rpm, 37 °C) for 12 h. After glutaraldehyde fixation and different concentrations of alcohol solutions 25 %, 50 %, 70 %, 80 %, 90 % and 100 % alcohol solution dehydration, the adhesion of bacteria on the surface was observed by SEM.

The *Chlorella* sp. and *Phaeodactylum Tricornutum* were selected to investigate the algal attachment on the coating surface. 50 mL of purchased algae was poured into 200 mL sterile artificial seawater, which was then added with 250 µL Guillard F/2 Medium to provide the necessary nutrients for algal growth, and then incubated in an light incubator (GXZ-380, 12 h light and 12 h dark) at 25 °C for one week. The blank glass, (Fe/TOB-PR)₁, (Fe/TOB-PR)₁₀ and (Fe/TOB-PR)₂₀ coatings (26 mm × 26 mm) were placed flat in 6-well plates with 10 mL of algae suspension and cultured at light incubator (12 h light and 12 h dark) at 25 °C, and ensured the algae concentration was above 10^8 cells per mL. Samples in *Chlorella* sp. solution were taken out at 24, 120 and 528 h, respectively, and in *Phaeodactylum Tricornutum* solution were taken out at 24 h and 120 h. Then all samples were washed gently-three times with artificial seawater. A biotype laser confocal microscope (LSCM, LEICA, TCS SP5, Germany) was used to record the algae fouling at six randomly selected locations of sample surface. Then, the area ratio of algae covered on the sample surface was calculated by using Photoshop (PS).

3. Results and discussion

3.1. Schiff base compound design and characterization

The Schiff base compound (TOB-PR) was synthesized via condensation polymerization from TOB and PR (Fig. 1A). The chemical structures of TOB, PR, and TOB-PR were identified based on FTIR, UV–vis (Fig. 2) and ¹H NMR spectrum (Fig. S1). As shown in Fig. 2a, the peaks at 1654 cm⁻¹ and 3230–3340 cm⁻¹ were corresponded to the —C=O and —OH stretching vibrations in PR. IR

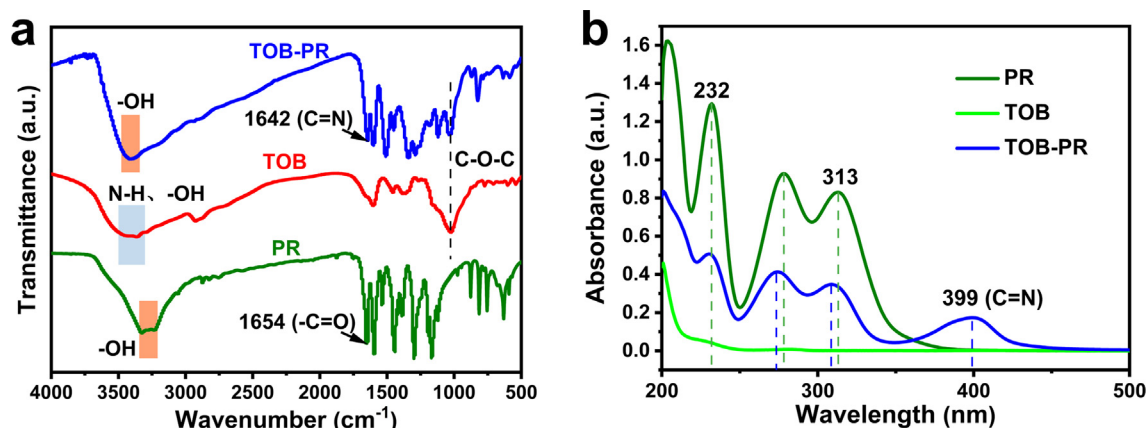


Fig. 2. Characterization of Schiff base compound. (a) FT-IR diagrams of TOB, PR, and TOB-PR; (b) UV-vis absorption spectra of TOB, PR, and TOB-PR.

spectrum of TOB also showed the characteristic peaks at 1022 cm⁻¹ (C—O—C) and 3343–3478 cm⁻¹ (N—H, —OH) [30]. Moreover, the synthesis of the Schiff base compound TOB-PR was confirmed by the disappearance of the characteristic —C=O peak and the clearly characteristic absorption of C=N (1642 cm⁻¹). This indicated that the —C=O groups reacted with the amino group of TOB to form the Schiff base characteristic bond C=N. Meanwhile, it was found that the peak height of ether bond was much lower than that in TOB after the synthesis of Schiff base TOB-PR, and the characteristic peak of —OH was slightly blue-shifted.

As demonstrated in UV-vis absorption spectra (Fig. 2b), PR appeared an obvious absorption peak at 200–400 nm, and the maximum absorption peak was 313 nm, which indicated that the molecular structure of PR contained a benzene ring structure. Furthermore, after the synthesis of the complex TOB-PR, the characteristic absorption band of the benzene ring undergone blue shift, and the intensity of catechol peak at 232 nm was significantly reduced. Moreover, a broad characteristic absorption peak of C=N was formed at 399 nm [31], which further indicating the TOB-PR was successfully synthesized. Further characterization of the molecular structure of TOB-PR by ¹H NMR was in Fig. S1 (Supporting Information). After condensation reaction, the characteristic peaks (δ 9.7 ppm, δ 2.5–2.8 ppm and δ 6.5–7.5 ppm) indicated that TOB-PR possessed the characteristic structure of both TOB and PR (Fig. S1c, Supporting Information), the successful synthesis of TOB-PR was further confirmed.

3.2. Construction of the (Fe/TOB-PR)_n coating

The assembly of the coating was divided into the following steps: a) TOB-PR as a catechol compound (as shown in Fig. 1A) mainly bonded to the glass substrate via hydrogen bonding [32]. b) Fixing Fe³⁺ on the surface group of catechol to obtain (Fe/TOB-PR)₁. c) Forming catechol-Fe³⁺-catechol bridge with catechol compound (as shown in Fig. 1B), repeated the first two steps to obtain (Fe/TOB-PR)_n coating (n = 1, 3, 5, 7, 10, 15, 20). As shown in Fig. 3a, the substrate changed from completely transparent to translucent brown with the increase of assembly times. At the same time, the UV spectra showed that the intensity of the characteristic peak of C=N increased continuously with the increase of assembly times, and there was only a small blue shift, which meant that the coating was successfully assembled and the chemical state of imine moieties was stable during the assembly process [33].

XPS full spectra showed that Fe/TOB-PR layers contained nitrogen and Fe³⁺, which were deposited on the glass surface (Fig. 3c). The peaks of N 1s and Fe 2p increased significantly, and the intensity of the substrate peak (Si 2p) completely disappeared after 10

layers of assembly. In addition, compared with TOB-PR on the glass, the O 1s spectra of (Fe/TOB-PR)₂₀ split a major peak (~531.1 eV), which could be assigned to Fe—O species (Fig. 3g) [34]. Fe—O arose from the coordination between TOB-PR and Fe³⁺. The peak at ~534.8 eV was the —CHO group in catechol compounds, and the peak increased with the number of assembled layers, indicating that the presence of some unreacted protocatechuic aldehyde in the Schiff base compound. Moreover, for the (Fe/TOB-PR)₂₀ coating, the Fe 2p_{3/2} signal showed a main peak at ~712 eV with a Fe 2p peak separation of ~13–14 eV, which was well consisted with the presence of Fe³⁺ species, it showed that the coordination bond between catechol and Fe³⁺ was established (Fig. 2g) [35,36].

Further, to identify the structure of the coating, we need to understand the number of ligand molecules around a central Fe³⁺. The TOB-PR and Fe³⁺ were mixed according to the ratio in Table S1 (Supporting information) to obtain Fe/TOB-PR solutions with different molar ratios. The complexation ratio between metal ions and catechol was observed by UV test. The complexation types of catechol and Fe³⁺ were 1:1, 2:1 and 3:1, and the greater the ratio was, the greater the peak intensity at 500–600 nm of UV spectrum was [19,22]. It could be seen from Fig. 3b that, when the TOB-PR:Fe³⁺ was 1:1.67 (Mass ratio), the characteristic peak was the highest due to complexation. Therefore, the ideal TOB-PR:Fe³⁺ complexation ratio was presumed to be 3:1 (Molar ratio). However, by integrating the sensitivity factors of each element peak in the (Fe/TOB-PR)₂₀ XPS spectra of Fig. 3c, the atomic composition of the coating was calculated, the ratio of TOB-PR:Fe³⁺ was around 1:1.25 (Table S2, Supporting information). This result showed that there might be three types of complexation between Fe³⁺ and catechol in the coating: 1:1, 1:2 and 1:3.

For self-renewing coatings containing anti-fouling agent, the surface was regularly eroded and renewed, while the antifoulant was released continuously from the matrix. Therefore, the thickness of coatings affected the service lifetime and anti-biological pollution performance of the coatings [7,8]. After adjusting the coating self-assembly steps and time for many times, the parameter conditions for preparing the coating were determined, and finally the ladder diagram and the increasing trend diagram of coating thickness for different layers were obtained (Fig. 4a and 4b). When the coating was assembled to the third layer, the thickness of (Fe/TOB-PR)₃ was about 9.4 nm, and the thickness of coating was about 104 nm at the 20th layer-(Fe/TOB-PR)₂₀. Different from the slow assembly speed of the first three layers, the thickness growth rate of each layer from the fourth layer was between 7 and 9 nm, which indicated that the combination of TOB-PR and matrix existed in the assembly process of the first three layers.

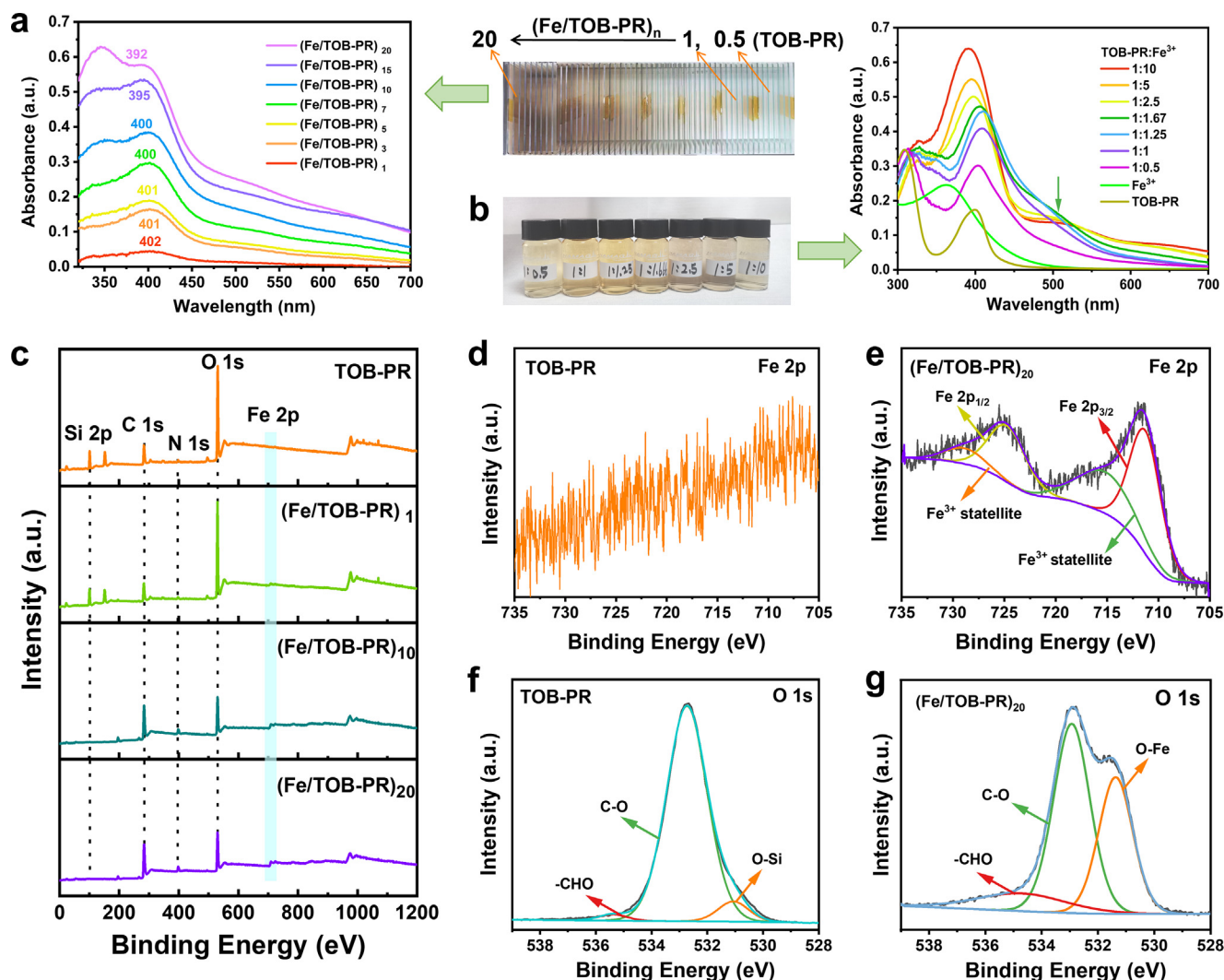


Fig. 3. The assembly of Schiff base compound and iron ion. (a) Photographs of $(\text{Fe}/\text{TOB-PR})_n$ coatings ($n = 1, 3, 5, 7, 10, 15, 20$) and their UV-vis absorption spectra; (b) Photographs of TOB-PR and Fe^{3+} with different molar ratio and their UV-vis absorption spectra; (c) XPS spectra for $(\text{Fe}/\text{TOB-PR})_n$ ($n = 0.5, 1, 10, 20$); Fe 2p core-level XPS spectra for (d) TOB-PR and (e) $(\text{Fe}/\text{TOB-PR})_{20}$; O 1s core-level XPS spectra for (f) TOB-PR and (g) $(\text{Fe}/\text{TOB-PR})_{20}$;

When the whole substrate was completely covered, the assembly speed of each layer was relatively accelerated, and the thickness growth rate tended to be stable. The surface morphology and roughness of $(\text{Fe}/\text{TOB-PR})_1$, $(\text{Fe}/\text{TOB-PR})_{10}$ and $(\text{Fe}/\text{TOB-PR})_{20}$ coatings were shown in Fig. 4d–f. The result exhibited that the surface roughness of $(\text{Fe}/\text{TOB-PR})_n$ coating achieved by layer-by-layer self-assembly showed little change with each additional layer, which indicated that each layer grew evenly. Meanwhile, according to the contact angle test results (Fig. 4c), the coating was hydrophilic and did not change greatly with the increase of thickness.

To verify the versatility of $(\text{Fe}/\text{TOB-PR})_{20}$ coating, we coated various planar substrates with different surface properties (hydrophilic, neutral, hydrophobic) - including glass, 316 stainless steel (Steel, Unpolished), polyamide (PA), ceramics, polytetrafluoroethylene (PTFE), polystyrene (PS), polydimethylsiloxane (PDMS) - with the $(\text{Fe}/\text{TOB-PR})_{20}$ coating. As shown in Fig. 4e, the color changed after the coating in all cases, demonstrating that the $(\text{Fe}/\text{TOB-PR})_{20}$ coating could be formed on a wide variety of substrates. However, the assembly effect of coatings on substrates with different surface characteristics was quite different. In Fig. 4f, the UV spectrum showed that the hydrophilic surface (Steel, PA, Ceramics, and Glass) was easier to assemble and the color was more uniform, while the hydrophobic surface

(polytetrafluoroethylene, PDMS) was prone to point assembly, resulting in the coating that couldn't cover the substrate evenly. The main principle of coating applicability was that catechol was combined with the substrate through hydrogen bond, metal catechol coordination, electrostatic interaction, cationic π interaction and other interactions on different substrates [37,38].

3.3. Degradation behaviors of the $(\text{Fe}/\text{TOB-PR})_{20}$ coating

To assess the degradation behaviors of Schiff base metal complex coatings, $(\text{Fe}/\text{TOB-PR})_{20}$ coatings were immersed in sterilized artificial seawater (Fig. 5) and deionized water (Fig. S3, Supporting Information), and the compositional changes in solutions and coatings were detected. In Fig. 5a and Fig. S3a (Supporting Information), the characteristic peaks of Fe^{3+} and benzene ring in catechol were around 350 nm, and the chelate peak of Fe^{3+} and catechol was at 500–600 nm. In both solutions, the intensity of the characteristic peak in the first 48 h increased relatively quickly, which indicated that the degradation rate of $(\text{Fe}/\text{TOB-PR})_{20}$ coating was relatively fast in the first 48 h. The increase of peak intensity after 48 h was obviously slowed down and uniform, indicating that the degradation was basically in equilibrium. The UV absorption intensity of coating (Fig. 5b and Fig. S3b, Supporting Information)

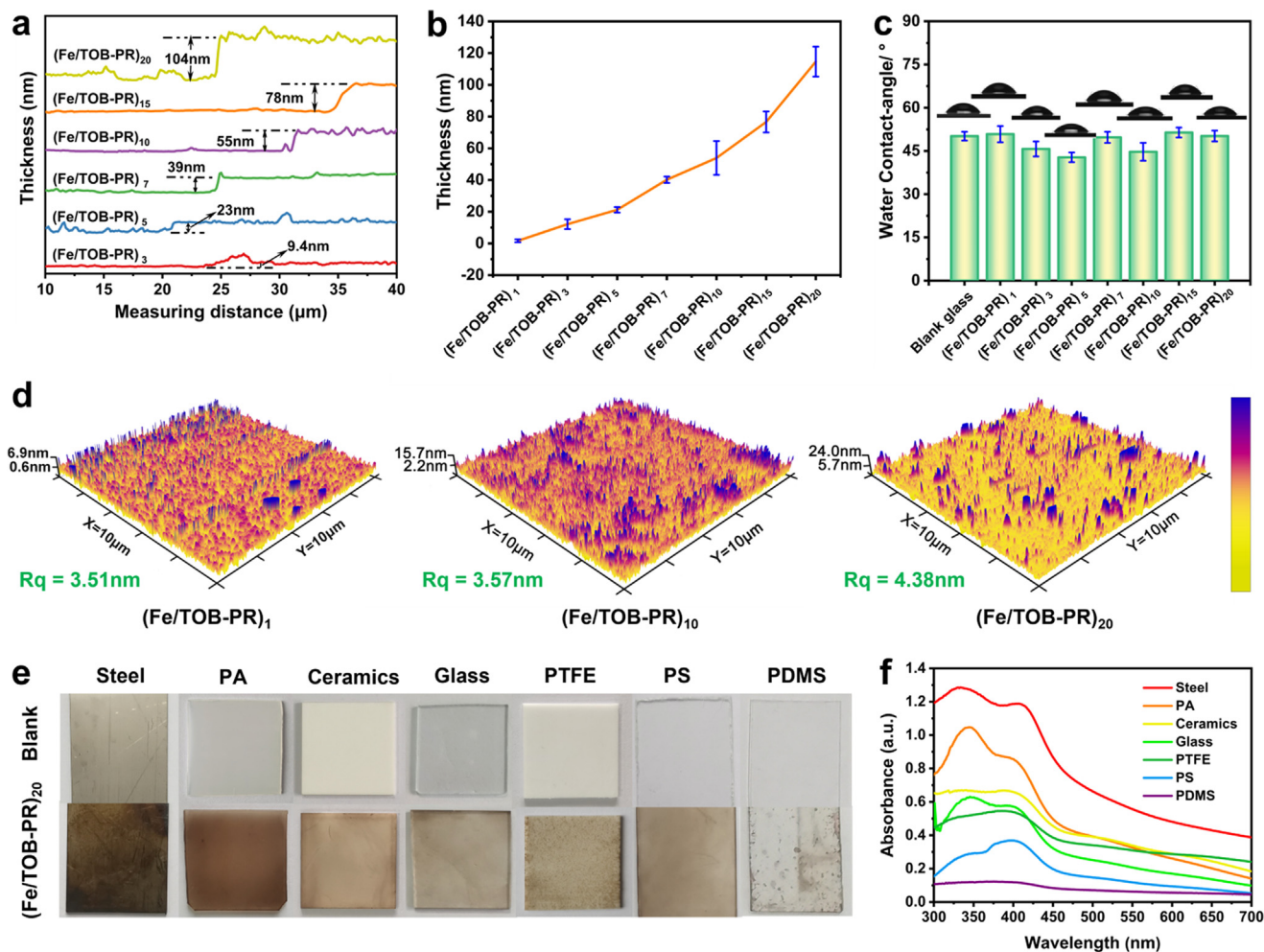


Fig. 4. Effect of assembly times on coating thickness, contact angle and roughness. (a) Thickness ladder diagram of (Fe/TOB-PR)_n coating; (b) Thickness of (Fe/TOB-PR)_n coating; (c) Water Contact Angles of (Fe/TOB-PR)_n coating; (d) SPM images of (Fe/TOB-PR)₁, (Fe/TOB-PR)₁₀ and (Fe/TOB-PR)₂₀; (e) Optical images of (Fe/TOB-PR)₂₀ coating assembled on different substrates; (f) UV-vis absorption spectra of (Fe/TOB-PR)_n coating assembled on different substrates.

also showed a slow decrease after 48 h. Comparing the coating thickness in the two solutions, the thickness change of the coating in the first 24 h in seawater was greater, indicated that it was in a state of slightly faster degradation state, while it maintained a more uniform thickness variation later. By comparing Fig. 5a and Fig. S3a, it was observed that the characteristic peak of chelate bond between catechol and Fe³⁺ was not obvious when degraded in the seawater, which might be due to that the pH value of seawater was between 7.9 and 8.4, which would accelerate the breakage of chelate bond [4]. When the coating was immersed in water, the Schiff base bond at the top was broken, and the interaction between catechol and Fe³⁺ was destroyed. At this time, TOB, PR, Fe³⁺ and some Schiff base metal chelates were present around the coating, forming an inert environment for biological survival. And the coating achieved self-renewal due to the removal of the old surface and the exposure of the new surface [33,39]. At the beginning of degradation, the roughness increased slightly, but the roughness changed little from 36 to 72 h, which might be due to that the degradation of coating surface kept a uniform rate after 36 h (Fig. S5, Supporting Information). Based on the experimental data above, it was speculated that the high-efficiency action time of the (Fe/TOB-PR)₂₀ antifouling coating was about 150 h. Due to the reversibility of the Schiff base characteristic bonds and the convenience of self-assembly, the coating was

expected to cyclic assembly to repeatedly realize short-term high-efficiency antifouling.

To further illustrate the self-renewal property of the Schiff base metal composite coating, we investigated the XPS of the (Fe/TOB-PR)₂₀ coating before and after 3 h and 72 h degradation (Fig. 5d-f, Fig. S6, Supporting Information). Firstly, after soaking for 3 h and 72 h, C, N fine spectra showed that the new surface after degradation showed the same chemical composition as the old surface before degradation. Both C 1s and N 1s showed that the C=N bond on the surface decreased after degradation, again demonstrating the breakage of the Schiff base bond [40]. Moreover, the comparative contents of aldehyde units (—CHO) at 535.30 eV and metal-oxygen bonding (O—Fe) at 531.34 eV had decreased pronouncedly, and amino units (—NH₂) at 403.87 eV had increased obviously, as proved by high resolution O 1s and N 1s XPS datum (Fig. 5e, f) [33,34,41]. This indicated that the content of aldehyde groups and metal-oxygen bonds had been decreased and amino groups on the coating surface have been increased. Further showed that the aldehyde groups and amine groups were formed after hydrolysis of Schiff Base. When the coating was degraded for 3 h, the aldehyde group (PR) was dissolved in water, while more amine groups remained on the surface. Therefore, it could be speculated that the interaction between amino groups and water could form a hydration layer on the coating interface [33,39].

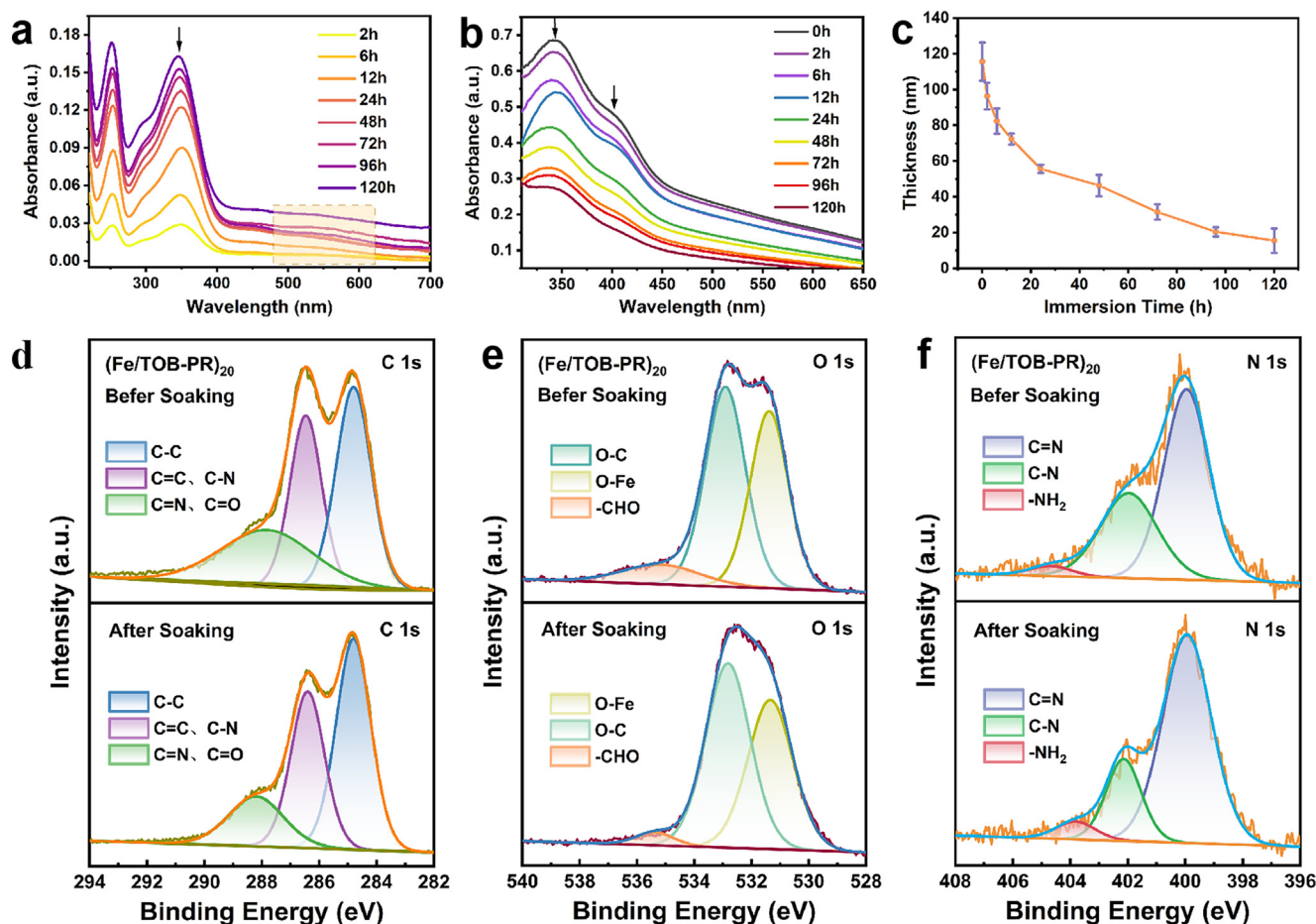


Fig. 5. Degradation behaviors of coatings and the resulting changes. (Fe/TOB-PR)₂₀ coating degrades in sterilized artificial seawater, the UV-vis absorption spectra of (a) the solution and (b) the coating; (c) Variation of coating thickness with immersion time in seawater; High-resolution (d) C 1s, (e) O 1s and (f) N 1s XPS results of (Fe/TOB-PR)₂₀ coating before and after 3 h immersion in seawater.

3.4. Antibacterial properties and mechanism

Antibacterial activity of TOB, PR, TOB-PR and Fe/TOB-PR against *E. coli* and *B. subtilis* (Fig. S8, Supporting Information) at the concentration of 1280 µg/mL was tested by ZOI assay. As shown in Fig. 6a–6c, except PR, all samples showed inhibition zones after incubation on agar plates for 24 h, indicating that the Schiff base compound TOB-PR and the Schiff base metal composite Fe/TOB-PR synthesized from TOB still displayed good bacteriostatic ability. TOB, TOB-PR and Fe/TOB-PR exhibited obvious inhibition zones with average diameters of about 24.5, 20.2 and 10.0 mm on agar plates, respectively. The area of the inhibition zone was ranked as TOB > TOB-PR > Fe/TOB-PR. TOB was the main substance that plays an antibacterial role. The Schiff base containing TOB molecular structure could be degraded to produce TOB only when it meets water. The degradation process was not instantaneous, which ensures the continuous release of antibacterial substances such as TOB. Therefore, the Fe/TOB-PR has a slow-release effect on the sustained and efficient antibacterial effect of the coating. The same test analysis was performed on Fe/TOB-PR coated samples (Transparent glass substrate) with different thicknesses (Fig. S7, Supporting Information). The results showed that the number of colonies in the place where Fe/TOB-PR samples assembled with one layer contacted the agar plate was evident less than the surrounding area. There was only a single colony at the (Fe/TOB-PR)₁₀ samples, while there was no colony appearing at the bottom of (Fe/TOB-PR)₂₀ samples. The results showed that the

(Fe/TOB-PR)₂₀ exhibited excellent antibacterial capability, and the coating thickness was also positively correlated with the antibacterial property.

The antibacterial behaviors of Schiff base metal composite coating (Fe/TOB-PR)_n (*n* = 1, 10, 20) were investigated by plate counting method with thickness of coating as independent variable, and Gram-negative *E. coli* and *B. subtilis* (Fig. S8, Supporting Information) were taken as model bacteria. For the accuracy of the experimental results, each sample was coated three times to obtain the average value. A group of macro-colony number pictures were selected (Fig. 6d). In addition, Fig. 6e and 6f showed the changes in bacteria number and bacteriostatic rate after coatings with different thickness were soaked in *E. coli* suspension for different time. Compared with the common glass control sample, the antibacterial effect of (Fe/TOB-PR)₁ coating was very limited, only about 50 % antibacterial rate after 24 h. In this process, the number of colonies increased slightly at 12 h (Fig. 6e). This may be due to the presence of a small amount of incompletely centrifuged nutrients in the bacterial suspension, leading to the reproduction and growth of incompletely killed bacteria. As time gone on, more Schiff bases ruptured on the coating surface to form TOB, which played a bactericidal effect, and the number of bacteria decreased again. The (Fe/TOB-PR)₁₀ and (Fe/TOB-PR)₂₀ coatings resulted in a significant reduction in the number of *E. coli* colonies after 18 h and 12 h of incubation, respectively, and the antibacterial rates of the two coatings reached 83 % and 97 % after 24 h. And from the curve change of antibacterial rate in Fig. 6f, it was found that

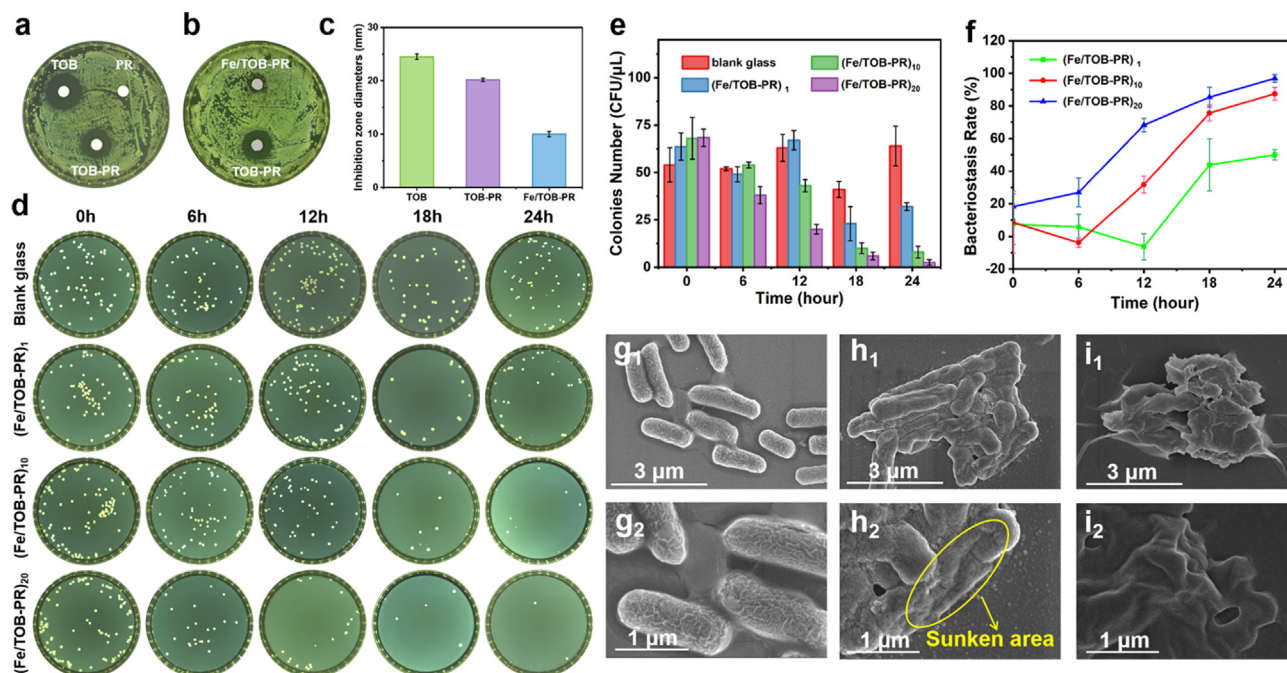


Fig. 6. Antibacterial properties and mechanism. (a–c) Zone of inhibition and diameters of inhibitory zones against gram-negative bacteria *E. coli* for TOB, PR, TOB-PR and Fe/TOB-PR; Colony distribution (d), colony count (e) and antibacterial rate (f) of blank glass, (Fe/TOB-PR)₁, (Fe/TOB-PR)₁₀ and (Fe/TOB-PR)₂₀ after immersed in *E. coli* suspension for 0, 6, 12, 18 and 24 h. The surface morphology of natural and treated *E. coli* (g) natural *E. coli*, (h) after being immersed in low concentration of TOB-PR solution, (i) after being immersed in high concentration of TOB-PR solution.

the antibacterial effect was more remarkable with the increase of the thickness of Fe/TOB-PR coating. It indicated that the thicker the Fe/TOB-PR coating was, the more natural antimicrobial agents were loaded, and the better the antimicrobial activity was. Furthermore, the above data showed that (Fe/TOB-PR)₂₀ coating has exhibited antibacterial properties, so the coating also could be applied to biomedical and related fields.

To better investigate the antibacterial mechanism of Fe/TOB-PR coating, Fig. 6g–i showed the surface morphology of natural *E. coli* as well as *E. coli* after being immersed in the low concentration of TOB-PR and high concentration of TOB-PR. The natural *E. coli* was intact and full in shape with no defects on the cell membrane. The cell membrane of *E. coli* after being immersed in the low concentration of TOB-PR was disrupted, and large sunken areas appeared on the surface, resulting in the outflow of intracellular components and death. However, compared with natural *E. coli*, the whole cells of *E. coli* after being immersed in the high concentration of TOB-PR were completely destroyed, and the cells were flaky with holes on the surface, which were more likely to agglomerate. This indicated after TOB-PR degradation, TOB sterilized *E. coli* cells by damaging their membranes, and even a low concentration of TOB could achieve good bactericidal effect [42,43].

3.5. Antifouling properties and mechanism

We evaluated antifouling performance of Fe/TOB-PR coatings by anti-bacterial and anti-algae adhesion experiments. The blank glass and (Fe/TOB-PR)₂₀ coating were shaken at a constant temperature for 12 h in the bacteria liquid cultured in LB, and the bacteria proliferated due to the abundant nutrients in the bacterial solution. Aggregated bacteria could be seen adhering to the blank glass. In contrast, only a small amount of nutrients from LB medium on the (Fe/TOB-PR)₂₀ coating, and no bacteria was observed (Fig. 7a and 7b). The main reason was that the Schiff base on the (Fe/TOB-PR)₂₀ surface was degraded due to imine hydrolysis reaction to generate the antibacterial substance TOB. The oscillating condi-

tion accelerated this process, which formed an inert environment for biological growth around the coating, resulting in the death of some bacteria and difficult adhesion. In other word, self-renewing Fe/TOB-PR coating showed a good synergistic effect of inhibiting bacteria and resisting material sedimentation in shaking condition.

Algae were one of the most important contributors for the formation of macrofouling. *Chlorella* and *Phaeodactylum tricornutum* are often selected by researchers to evaluate the antifouling ability of materials [30]. The two species of algae were selected in this fouling test, and the antifouling performance of the coating was further evaluated by monitoring the algal adhesion area on the coating surface under static conditions (Fig. 7e, 7f). Fig. 7c and 7d showed the adhesion of *Phaeodactylum tricornutum* (red) and *Chlorella* sp. (green), respectively, taken by fluorescence microscopy. The coverage rate of *Phaeodactylum tricornutum* on the glass substrate reached 10 % in 24 h (Fig. 7g), and the surface covered area reached 37.4 % after 120 h. However, after 120 h, the amount of *Phaeodactylum tricornutum* adhering to Fe/TOB-PR coating decreased obviously. And the thicker the coating was, the lesser algae adhered to the surface was. Compared with *Phaeodactylum tricornutum*, *Chlorella* sp. was not easy to adhere to substrate because of its spherical shape, so there was no significant difference between blank glass and coatings after 120 h (Fig. 7d). However, after 528 h, compared with the blank glass, the adhesion degree of algae on the surface of (Fe/TOB-PR)₂₀ coating decreased obviously (Fig. 7h). To sum up, the Schiff base metal complex coating had a certain static antifouling ability under static and shaking conditions, and the thicker the coating was, the better the antifouling performance was.

The possible antifouling mechanism of Schiff base metal composite coating could be summarized as follows: (I) With the continuous degradation of Schiff base, TOB was released continuously, which formed an inert environment for biological growth around the coating and inhibited the formation of primary fouling; (II) The outermost layer of the coating was constantly self-

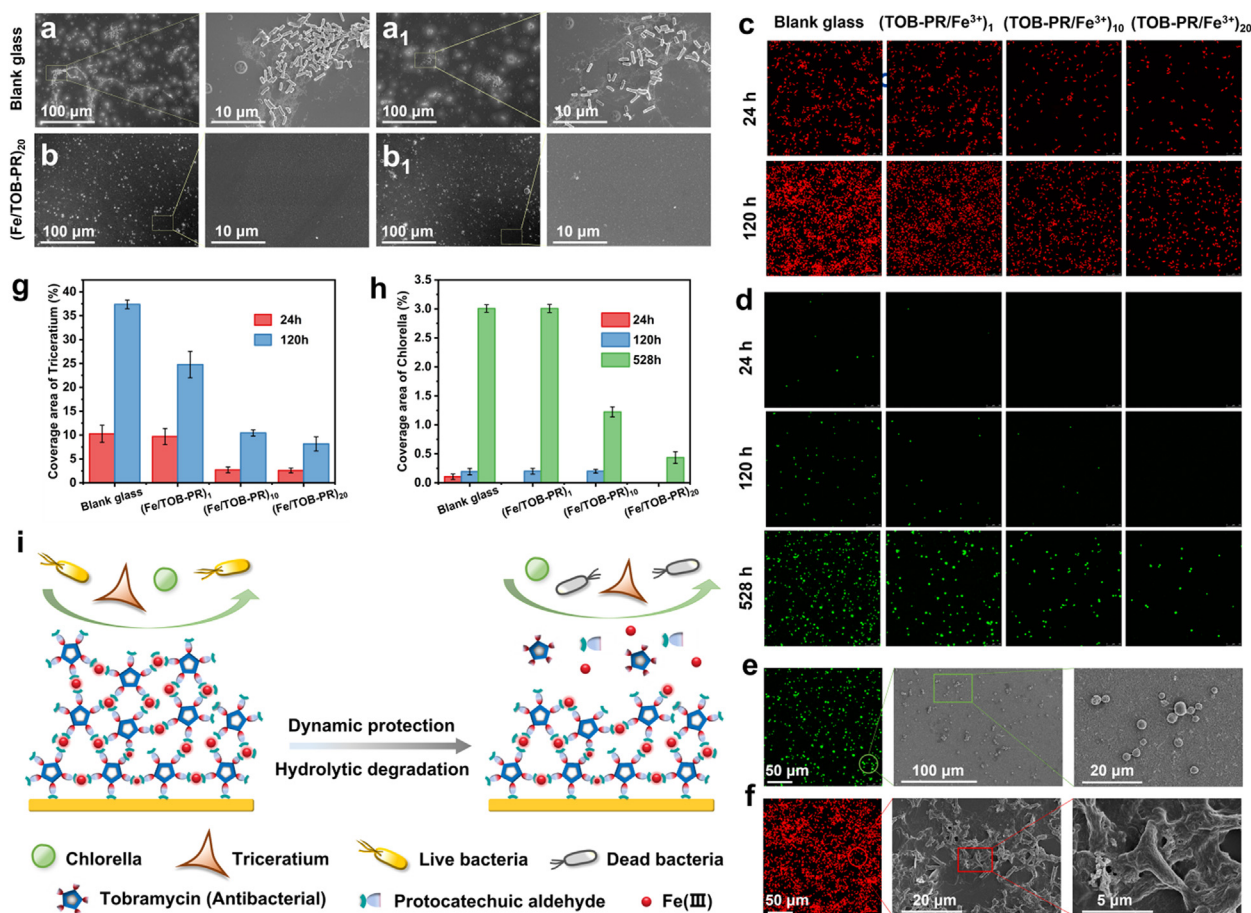


Fig. 7. Antifouling properties and mechanism. SEM images of samples immersed in the bacteria liquid (LB) for 12 h: (a, a₁) Blank glass, (b, b₁) (Fe/TOB-PR)₂₀ coating; Laser confocal photograph of the sample surface: (c) *Phaeodactylum tricornutum* and (d) *Chlorella* sp. The spots represented diatom adhering to the surface; (e) SEM images of *Chlorella* sp. adhered on blank glass after 528 h; (f) SEM images of *Phaeodactylum tricornutum* adhered on blank glass after 120 h; Percent of the coverage area of *Phaeodactylum tricornutum* (g) and *Chlorella* sp. (h) on different coatings; (i) Schematic diagram of the antifouling mechanism of (Fe/TOB-PR)₂₀ coating.

renewal by degradation, preventing fouling organisms from adhering, whether under static or shaking conditions; (III) The imine bond of the schiff base decomposed to produce aldehyde ends or amino that were exposed to the surface of the coating, and aldehyde ends or amino could form hydrogen bonds with water, which was speculated to form a hydrated layer on the anti-fouling interface. While marine organisms may need to break this hydration layer barrier in order to adhere firmly to the material interface [33,44–46], thereby reducing the likelihood of adhesion. In addition, importantly, the coating degradation products were all from natural, so the products were environment-friendly and good biocompatible, and microplastics would not be accumulated in seawater due to the low molecular weight of the product. In the future, we will search more natural products of aldehydes and amines in nature, improve the structure of the coating by constructing and optimizing the preparation process of Schiff base coating, and understand more deeply the degradation law and anti-fouling mechanism of the material, in order to achieve long-term antibacterial and anti-biological fouling.

4. Conclusions

In summary, we explored a new environment-friendly dual-function anti-fouling strategy that simultaneous self-renewal and sterilization. Schiff base compounds with dynamic imine covalent bonds was synthesized from the natural extract tobramycin

(TOB) and protocatechuic aldehyde (PR). Furthermore, a Schiff base metal composite coating (Fe/TOB-PR)_n with TOB as the antifouling functional monomer was prepared by a layer-by-layer self-assembly method. And the suitability of coating for a wide range of substrates was also explored. In terms of degradation, the coating showed regular and controllable degradation. On the one hand, as the degradation of the coating, the natural antibacterial agents were released into the water, which could effectually improve the antibacterial properties and the antibacterial rate reached 97 % in 24 h. On the other hand, due to the surface renewal caused by degradation, the coating showed anti-biofouling effect to a certain extent under both static and shaking conditions. In other words, (Fe/TOB-PR)₂₀ coating exhibited excellent antimicrobial properties and good resistance to microbial adhesion via the synergistic effect of attack and defense. Eventually, the coating completely degraded to natural micromolecule monomer, avoiding the occurrence of microplastics. This environmental friendly coating showed promising application in the field of marine antifouling.

CRediT authorship contribution statement

Wenjuan Liu: Methodology, Validation, Formal analysis, Investigation, Writing – original draft. **Minglong Yan:** Conceptualization, Validation, Supervision. **Wenjie Zhao:** Conceptualization, Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (52103133), Zhejiang Province Key Technology Project (2022C01183&2022C01213), Ningbo Natural Science Foundation (2021J211), the Science and Technology Innovation 2025 Major Project of Ningbo (No. 2020Z053) and the Director Fund of Ningbo Institute of Materials Technology and Engineering (No. E20301QF08).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2022.08.178>.

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